

Skills Summary

Halves and Fourths — Making Equal Shares

Use this reference while completing your worksheet or when studying.

Skill 1 — Checking whether two parts are really halves

The rule:

Two parts are halves only when the two parts hold the SAME number of squares.

- Cutting a shape into two pieces does *not* make halves by itself.
- Count the squares in each part, then compare the two counts.
- Same count → halves. Different count → not halves.

To know what a fair half should be, share the whole count between the two parts.

Example: A strip of 8 squares is cut into 2 parts. How many squares belongs in each half?

Step 1. A half must hold the same count as the other half, so share the whole strip equally between the two parts.

Step 2. Share 8 squares between 2 parts: 4 for one part and 4 for the other.

Each half holds 4 squares. A part with any other count is not a half.

Try it: A strip of 10 squares is cut into 2 equal parts. How many squares are in each half?

check: 5

Skill 2 — Fixing a cut that is not halves

How to fix an unequal cut:

- Put the two counts together to get the whole.
- Share the whole between the two parts. That is the **fair share**.
- Count up from the small part to the fair share. That is how many squares move across.

Watch the size of the move: moving the whole gap makes the small part the big one. Only *half* the gap moves.

Example: One part holds 3 squares and the other holds 9. How many squares move so both parts are halves?

Step 1. Halves must match, so find the fair share first, then see how far the small part is from it.

Step 2. The two parts together hold 12 squares, and shared between two parts the fair share is 6 each. Count up from 3 to 6.

The move is 3 squares. Check: $3 + 3 = 6$ and $9 - 3 = 6$ ✓

Try it: One part holds 2 squares and the other holds 10. How many squares move so both parts are halves?

check: 4

Skill 3 — Fourths: all four parts must match

The rule: Fourths are four parts that all hold the SAME number of squares.

- Four pieces of any size are just four pieces, not fourths.
- Fold once and you have halves. Fold the halves again and you have fourths.
- Share the whole count into four equal groups to find one fourth.

Example: A rectangle of 12 squares is cut into fourths. How many squares are in one fourth?

Step 1. Fourths are four matching parts, so share the whole count into 4 equal groups.

Step 2. Share 12 squares into 4 groups: 3, 3, 3, 3.

One fourth holds **3** squares.

Try it: A rectangle of 16 squares is cut into fourths. How many squares are in one fourth?

check: 4

(!) Watch out: The name tells you how many *equal* parts, not just how many cuts. Two parts that do not match are not halves, and four parts that do not match are not fourths — so always count both (or all four) parts before you use the name.